

Measurement uncertainty budget – GUM H.1 – End-gauge calibration (worked example)

Laboratory: Worked example

Prepared by: —

Date: 2026-10-08

Procedure: JCGM 100:2008 Annex H.1

Equipment: Comparator, standard end gauge

Conditions: Bed temperature (19.9 ± 0.5) °C

Measurand and measurement model

Length of end gauge at 20 °C (l , mm). Nominal 50 mm end gauge calibrated by comparison with a standard of the same nominal length (JCGM 100:2008, H.1).

Model: $l = l_S + d - l_S(\alpha + \alpha_S \theta)$. Sensitivity coefficients marked “model” are the partial derivatives $\partial f / \partial x_i$ evaluated numerically at the input estimates (GUM 5.1.3).

Variable	Estimate	Unit	Description
l_S	50.000623	mm	Length of the standard at 20 °C (certificate)
d	2.15e-4	mm	Mean measured difference (5 readings) = 215 nm
α_S	1.15e-5	1/°C	Thermal expansion coefficient of standard
θ	-0.1	°C	Deviation of bed temperature from 20 °C
α	0	1/°C	Difference in expansion coefficients
$d\theta$	0	°C	Difference in temperature of the gauges

Uncertainty budget

#	Component	Source of data	Type	Distribution	Input value	Divisor	$u(x_i)$	c_i	$c_i \cdot u(x_i)$ [mm]	v_i	% contrib.
1	Calibration of the standard end gauge $u(l_S)$	Calibration certificate of the reference standard Certificate: $U = 0.075$ μm , $k = 3$, $v_{\text{eff}} = 18$ (GUM H.1.3.1)	B	Normal (U, k)	7.5e-5 mm	$k = 3$	2.5e-5 mm	1	2.5e-5	18	62.4 %
2	Repeated observations of the difference $u(d)$	Historical / pooled data Pooled $s = 13$ nm from 25 observations; mean of 5 readings (GUM H.1.3.2)	A	Type A (pooled)	$s_p = 1.3\text{e-}5$, $m = 5$ mm	$\sqrt{5}$	5.814e-6 mm	1	5.814e-6	24	3.4 %

#	Component	Source of data	Type	Distribution	Input value	Divisor	$u(x_i)$	c_i	$c_i \cdot u(x_i)$ [mm]	v_i	% contrib.
3	Comparator – random effects $u(d1)$	Calibration certificate of the reference standard Comparator certificate: $\pm 0.01 \mu\text{m}$ at 95 %, 6 replicates ($v = 5$)	B	Normal (U, p %)	1e-5 mm	$t(95 \%, v=5) = 2.571$	3.89e-6 mm	1	3.89e-6	5	1.5 %
4	Comparator – systematic effects $u(d2)$	Calibration certificate of the reference standard Comparator certificate: $0.02 \mu\text{m}$ at "three sigma"; reliable to 25 % - $> v = 8$	B	Normal (U, k)	2e-5 mm	$k = 3$	6.667e-6 mm	1	6.667e-6	8	4.4 %
5	Thermal expansion coefficient of standard $u(\alpha S)$	Reference data / handbooks $\alpha S = 11.5\text{e-}6 / ^\circ\text{C}$, rectangular $\pm 2\text{e-}6 / ^\circ\text{C}$ (GUM H.1.3.3)	B	Rectangular	2e-6 1/°C	$\sqrt{3}$	1.155e-6 1/°C	0	0	∞	0.0 %
6	Mean temperature of bed $u(\bar{\theta})$	Environmental conditions Reported (19.9 ± 0.5) °C; $u(\bar{\theta}) = 0.2$ °C (GUM H.1.3.4)	B	Standard uncertainty	0.2 °C	1	0.2 °C	0	0	∞	0.0 %
7	Cyclic variation of room temperature $u(\Delta)$	Environmental conditions Cyclic ± 0.5 °C, U-shaped (arcsine) (GUM H.1.3.4)	B	U-shaped	0.5 °C	$\sqrt{2}$	0.3536 °C	0	0	∞	0.0 %
8	Difference in expansion coefficients $u(\delta\alpha)$	Reference data / handbooks Bounds $\pm 1\text{e-}6 / ^\circ\text{C}$ rectangular, reliable to 10 % -> $v = 50$ (GUM H.1.3.5)	B	Rectangular	1e-6 1/°C	$\sqrt{3}$	5.774e-7 1/°C	5.0001	2.887e-6	50.0	0.8 %

#	Component	Source of data	Type	Distribution	Input value	Divisor	$u(x_i)$	c_i	$c_i \cdot u(x_i)$ [mm]	v_i	% contrib.
9	Difference in temperature of gauges $u(\delta\theta)$	Environmental conditions Bounds ± 0.05 °C rectangular, reliable to 50 % $\rightarrow v = 2$ (GUM H.1.3.6)	B	Rectangular	0.05 °C	$\sqrt{3}$	0.02887 °C	-5.7501e-4	-1.66e-5	2	27.5 %

Results

Estimate of the measurand y	50.000838 mm
Combined standard uncertainty $u_c(y)$	3.1658e-5 mm
Relative combined standard uncertainty $u_c/ y $	6.33e-7
Effective degrees of freedom v_{eff} (Welch–Satterthwaite)	16.7
Coverage	$p = 99\%$, v used = 16 (truncated, GUM G.6.4)
Coverage factor k	2.921
Expanded uncertainty $U = k \cdot u_c$	9.2467e-5 mm

$I = (50.000838 \pm 0.000092) \text{ mm}$

$I = (50.000838 \pm 0.000092) \text{ mm}$. The reported expanded uncertainty $U = k \cdot u_c$ is based on a combined standard uncertainty $u_c = 0.000032 \text{ mm}$ multiplied by a coverage factor $k = 2.921$ from the t-distribution for $v = 16$ ($v_{\text{eff}} = 16.7$), level of confidence approximately 99 %, defining an interval estimated to have a level of confidence of approximately 99 %.

Notes and warnings

- Thermal expansion coefficient of standard has sensitivity $c = 0$ and contributes nothing to first order (consider second-order terms, GUM 5.1.2 note, H.1.7).
- Mean temperature of bed has sensitivity $c = 0$ and contributes nothing to first order (consider second-order terms, GUM 5.1.2 note, H.1.7).
- Cyclic variation of room temperature has sensitivity $c = 0$ and contributes nothing to first order (consider second-order terms, GUM 5.1.2 note, H.1.7).

Notes

Published result: $u_c = 32 \text{ nm}$, $v_{\text{eff}} = 16$, $k = 2.92$, $U_{99} = 93 \text{ nm}$ (GUM multiplies the rounded $u_c = 32 \text{ nm}$ by 2.92).

Method

Uncertainty evaluated per JCGM 100:2008 (GUM) and NIST TN 1297: standard uncertainties from Type A (statistical) and Type B evaluations, combined by the law of propagation of uncertainty (first-order Taylor series, GUM 5.1.2 / 5.2.2), effective degrees of freedom by the Welch–Satterthwaite formula (GUM G.4.1), coverage factor

from the Student t-distribution (GUM G.3/G.6.4), and reported to two significant digits (GUM 7.2.6). Rounding: rounded to nearest.

References

1. JCGM 100:2008 (GUM 1995 with minor corrections), Evaluation of measurement data – Guide to the expression of uncertainty in measurement.
2. NIST Technical Note 1297 (1994), B. N. Taylor & C. E. Kuyatt, Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results.
3. ISO/IEC 17025:2017, General requirements for the competence of testing and calibration laboratories, clause 7.6 (evaluation of measurement uncertainty) and 7.8.4 (calibration certificates).
4. ILAC P14:09/2020, ILAC Policy for Measurement Uncertainty in Calibration.
5. EA-4/02 M:2022, Evaluation of the Uncertainty of Measurement in Calibration.

Prepared by / date

Reviewed / approved by / date

Generated by Uncertainty Budget Calculator Made Simple on 2026-10-08 23:32 UTC. The laboratory remains responsible for the validity of the model, inputs and their sources (ISO/IEC 17025 7.6).